



< Previous

✓✓

Next >

9.2.3 More properties of eigenvalues and vectors

 Bookmark this page

Pursue a verified certificate



Verified Certificate

THE UNIVERSITY of TEXAS SYSTEM

This is to certify that

Sandipan Dey

successfully completed and received a passing grade in

UT.ALA:Advanced Linear Algebra: Foundations to Frontiers

a course of study offered by UTAustinX, an online learning initiative of University of Texas System.

**Maurilio Pugliesi**
Professor
UniversityX

**Justine Doe, Ph.D.**
Professor
UniversityX

Verified Certificate

Valid Certificate ID

Issued Month / Year

12345678901234567890

- Unlimited access to all course materials
- Shareable [verified certificate](#)
- AI learning assistant and translations

Upgrade for \$99

Reading assignment

1/1 point (graded)



Read Unit 9.2.3 of the notes. [\[LINK\]](#)

☒ done



Submit

✓ Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 09 / 09.2.3

Add a Post

Show all posts ▼

by recent activity ▼

There are no posts in this topic yet.

✕

Homework 9.2.3.1

1/1 point (graded)

Let λ be an eigenvalue of $A \in \mathbb{C}^{m \times m}$ and let

$$\mathcal{E}_\lambda(A) = \{x \in \mathbb{C}^m \mid Ax = \lambda x\}.$$

be the set of all eigenvectors of A associated with λ plus the zero vector (which is not considered an eigenvector). Show that $\mathcal{E}_\lambda(A)$ is a subspace.

☒ done



Solution:

For a complete solution see the [notes](#).

Submit

Answers are displayed within the problem

Homework 9.2.3.2

1/1 point (graded)

Let $D \in \mathbb{C}^{m \times m}$ be a diagonal matrix. Give all eigenvalues of D . For each eigenvalue, give a convenient eigenvector.

☒ done



Solution:

For a complete solution see the [notes](#).

Submit

Answers are displayed within the problem

Homework 9.2.3.3

1/1 point (graded)
Compute the eigenvalues and corresponding eigenvectors of

$$A = \begin{pmatrix} -2 & 3 & -7 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

(Recall: the solution is not unique.)

☒ done



Submit

Correct (1/1 point)

Homework 9.2.3.3

1/1 point (graded)
Let $U \in \mathbb{C}^{m \times m}$ be an upper triangular matrix. Give all eigenvalues of U . For each eigenvalue, give a convenient eigenvector.

☒ done



Solution:

For a complete solution see the [notes](#).

Submit

Answers are displayed within the problem



edX

- [About](#)
- [Affiliates](#)
- [edX for Business](#)
- [Open edX](#)
- [Careers](#)
- [News](#)

Legal

- [Terms of Service & Honor Code](#)
- [Privacy Policy](#)
- [Accessibility Policy](#)
- [Trademark Policy](#)
- [Sitemap](#)
- [Cookie Policy](#)
- [Your Privacy Choices](#)

Connect

- [Idea Hub](#)
- [Contact Us](#)
- [Help Center](#)
- [Security](#)
- [Media Kit](#)

